

UCP Document

NexusNotes

Centralized Academic Resource Platform

Project Team: Code Club

Ramasani Adi Pranav – 24CSB0A57

Syed Mohammad Sami – 24CSB0A77

Janapala Harsha Vardhan Reddy – 24CSB0A28

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1 Introduction

This document outlines the use cases for the Nexus Notes learning platform and calculates the estimated software size using the Use Case Points (UCP) methodology developed by Gustav Karner.

2 Use Case Inventory

Based on the system requirements, the following use cases and actors have been identified. The weight of each use case is determined by its number of transactions: Simple (1 to 3), Average (4 to 7), or Complex (8 or more).

ID	Use Case Name	Complexity
UC-01	Upload Academic Resource	Complex
UC-02	Search and Download Resource	Average
UC-03	View Tech-Trend Feed	Average
UC-04	Rate and Review Resource	Average
UC-05	User Management	Average
UC-06	Content Moderation	Average
UC-07	System Monitoring	Simple
UC-08	Create Personalized Learning Path	Average
UC-09	Bookmark Resources	Simple
UC-10	Generate Analytics Report	Simple

Table 1: Use Cases for Nexus Notes

3 UCP Calculation Parameters

3.1 Unadjusted Actor Weight (UAW)

Actors are classified as Simple (API), Average (Protocol), or Complex (GUI).

- **Complex Actors (Weight 3):** Student, Administrator (2 actors)
- **Average Actors (Weight 2):** None (0 actors)
- **Simple Actors (Weight 1):** External Tech API (1 actor)

UAW Calculation:

$$(2 \times 3) + (0 \times 2) + (1 \times 1) = 7$$

3.2 Unadjusted Use Case Weight (UUCW)

- **Complex Use Cases (Weight 15):** 1
- **Average Use Cases (Weight 10):** 6

- **Simple Use Cases (Weight 5):** 3

UUCW Calculation:

$$(1 \times 15) + (6 \times 10) + (3 \times 5) = 90$$

3.3 Complexity Factors

Using the evaluated baseline values for the technical and environmental complexities of the Nexus Notes platform:

- **Total Technical Factor (TF):** 45.5
- **Technical Complexity Factor (TCF):** $0.6 + (45.5/100) = 1.055$
- **Total Environmental Factor (EF):** 19.5
- **Environmental Complexity Factor (ECF):** $1.4 + (-0.03 \times 19.5) = 0.815$

4 Final Software Size Estimation

The UCP is calculated based on the following formula:

$$UCP = (UUCW + UAW) \times TCF \times ECF$$

$$UCP = (90 + 7) \times 1.055 \times 0.815$$

$$\mathbf{UCP = 83.40}$$

Assuming 28 man-hours per use case point will be used, the Estimated Effort for the Nexus Notes platform is approximately:

$$\mathbf{2,335 \text{ Hours}}$$

5 Appendix: C++ Program for UCP Calculation

The following C++ program automates the calculation of the Use Case Points based on the formulas provided above.

```
1 #include <iostream>
2 #include <iomanip>
3
4 using namespace std;
5
6 int main() {
7     // 1. Unadjusted Actor Weight (UAW)
8     int simpleActors, averageActors, complexActors;
9     cout << "Enter number of Simple Actors (API): ";
10    cin >> simpleActors;
11    cout << "Enter number of Average Actors (Protocol): ";
12    cin >> averageActors;
13    cout << "Enter number of Complex Actors (GUI): ";
14    cin >> complexActors;
15
16    // Weightings: Simple=1, Average=2, Complex=3
17    double uaw = (simpleActors * 1) + (averageActors * 2) + (
        complexActors * 3);
18
19    // 2. Unadjusted Use Case Weight (UUCW)
20    int simpleUC, averageUC, complexUC;
21    cout << "Enter number of Simple Use Cases (1-3 transactions):
        ";
22    cin >> simpleUC;
23    cout << "Enter number of Average Use Cases (4-7 transactions)
        : ";
24    cin >> averageUC;
25    cout << "Enter number of Complex Use Cases (8+ transactions):
        ";
26    cin >> complexUC;
27
28    // Weightings: Simple=5, Average=10, Complex=15
29    double uucw = (simpleUC * 5) + (averageUC * 10) + (complexUC
        * 15);
30
31    // 3. Technical Complexity Factor (TCF)
32    double tf;
33    cout << "Enter Total Technical Factor (TF): ";
34    cin >> tf;
35    double tcf = 0.6 + (tf / 100.0);
36
37    // 4. Environmental Complexity Factor (ECF)
38    double ef;
39    cout << "Enter Total Environmental Factor (EF): ";
40    cin >> ef;
41    double ecf = 1.4 + (-0.03 * ef);
```

```

42
43 // 5. Calculate Final UCP
44 double ucp = (uucw + uaw) * tcf * ecf;
45
46 // Output Results
47 cout << "\n--- UCP Calculation Results ---\n";
48 cout << "UAW: " << uaw << endl;
49 cout << "UUCW: " << uucw << endl;
50 cout << "TCF: " << fixed << setprecision(3) << tcf << endl;
51 cout << "ECF: " << fixed << setprecision(3) << ecf << endl;
52 cout << "Final Use Case Points (UCP): " << fixed <<
    setprecision(2) << ucp << endl;
53
54 // Effort Estimation
55 cout << "Estimated Effort (Assuming 28 hrs/UCP): "
56     << fixed << setprecision(0) << (ucp * 28)
57     << " Hours" << endl;
58
59 return 0;
60 }

```